

TECHNICAL REPORT

UNLOCKING BEHAVIORAL INTELLIGENCE IN AIRLINE LOYALTY PROGRAMS

Churn Prediction, Customer Segmentation, and a Smart Retention Strategy for the Loyalty Program

This report analyzes the behavior of 16,737 Canadian airline loyalty members, builds a model to identify members at risk of disengaging, segments the customer base into actionable groups, and translates those findings into specific retention actions. It is written for a business audience — a marketing or finance leader — rather than a technical reviewer

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AGENDA

“This agenda outlines the key topics covered in this report, providing a clear structure to support discussion and review.”

- **Executive Summary**
- **How We Framed the Problem**
- **Data Overview and Cleaning Decisions**
- **Churn Prediction**
- **Customer Segmentation and Value**
- **Smart Retention**
- **The Most Interesting Things We Found**
- **Recommendations**
- **Working Prototype**
- **Assumptions, Limitations, and Future Work**

EXECUTIVE SUMMARY

The airline's loyalty program currently relies on Customer Lifetime Value (CLV) and formal cancellation records to understand its members. This analysis shows that neither is sufficient on its own. CLV, as currently calculated, is almost identical across customers who never fly and customers who fly over twenty times a year — it does not reflect how engaged a member actually is. Formal cancellations, meanwhile, capture only a fraction of the members who are quietly disengaging.

To address this, we built three connected pieces of work:

1. A churn prediction model that flags members likely to disengage in the next year, using only information available at the time of prediction (no future data leakage). The model achieves a strong 0.89 ROC-AUC, and its single biggest predictor is recency — how recently a member last flew.

2. A segmentation framework that groups all 16,737 members into five behaviorally distinct, business-actionable personas — ranging from “Loyal Frequent Flyers” to a large “Lapsed / Critical Risk” group with a 64% churn rate.

3. A retention action matrix that converts each segment into a specific, ready-to-execute campaign — who receives it, when, why, and in what form — surfaced through a working dashboard prototype.

The report closes with three recommendations that a CFO and a CMO can act on jointly: redefining how customer value is measured, launching an urgent win-back program for the highest-risk segment, and activating a large pool of unused loyalty points before it becomes a silent liability.

2. HOW WE FRAMED THE PROBLEM

The brief asked us to define churn ourselves, since formal cancellation is only one form of disengagement. We adopted a two-part definition:

- A member is considered “churned” if they formally cancelled their loyalty membership in 2018, OR
- They remained enrolled but recorded zero flights in the second half of 2018 (July–December) — a strong signal of quiet disengagement, even without a formal exit.

This definition produced a churn rate of 16.3% across the customer base — a realistic, modelable rate that is neither vanishingly rare nor so common that prediction becomes trivial.

2.1 Avoiding Data Leakage

A core requirement of this brief was that the model must only use information that would genuinely have been available at the time of prediction. While the loyalty history spans enrollments from 2012 to 2018, the monthly flight activity data — our main behavioral signal — only covers January 2017 to December 2018, a 24-month window.

We split this window cleanly in two:

- Feature window (Jan–Dec 2017): all predictive features — flight frequency, distance, points activity, seasonality, recency — are built only from this period.
- Outcome window (Jan–Dec 2018): used only to define whether a member churned. None of this data is used as a model input.

This guarantees the model is making a genuine forward-looking prediction, exactly as it would need to in production: “given what we know about this member through the end of 2017, will they disengage during 2018?”

3. DATA OVERVIEW AND CLEANING DECISIONS

The dataset combines four files: a loyalty history table (one row per customer, with demographics, CLV, and cancellation records), a monthly flight activity table, a calendar dimension, and a data dictionary. Below are the key data quality issues we found and how we resolved them — each reflects a deliberate judgment call rather than a default “clean and move on” approach.

3.1 Missing Salary (25% of members)

A quarter of members had no recorded salary. Dropping them would have removed a meaningful share of the customer base, so we imputed missing values using the median salary within each education group, and added a “Salary Missing” flag so the model can learn whether the absence itself is informative (for example, it may correlate with certain enrollment types).

3.3 New 2018 Enrollees Are Not “At Risk” — They're New

3,010 members (18% of the base) enrolled during 2018, meaning they have no 2017 flight history at all. If left in the churn model, their all-zero 2017 features would look identical to a long-standing member who suddenly went silent — but the underlying story (and the right marketing response) is completely different. We separated these members into their own group, used for segmentation and onboarding strategy but excluded from the churn model's training and evaluation.

3.5 Members Who “Cancelled” but Still Show Later Flight Activity

Around 2,067 members have a recorded cancellation date, all logically after their enrollment date (no member cancels before joining). However, some members with cancellation dates as early as 2013–2016 still show flight activity in 2017–2018. We flag this as an open question rather than silently resolving it: it may mean “cancellation” refers to a loyalty tier or status change rather than a full program exit, or it may be a quirk of how this dataset was constructed. We recommend the data/IT team clarify the exact meaning of this field, since it affects how “churn” should ultimately be defined in production.

3.2 Flight Activity Window Is Shorter Than Expected

Although members enrolled between 2012 and 2018, the flight activity records only cover 2017–2018. This was the single most consequential finding in the cleaning phase, because it directly defined how churn could be modeled — see Section 2.1. We treat this as a structural property of the dataset and built the entire prediction framework around it, rather than attempting to extrapolate behavior for years with no data.

3.4 Points Redeemed Exceeding Points Accumulated in a Single Month

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3.6 Columns Dropped

“Country” contains a single value (Canada) across all records and was dropped as uninformative. “City” and “Postal Code” were dropped due to high cardinality that would not generalize to new members. “Cancellation Year/Month” were dropped from the model's input features, since they directly encode the outcome we are predicting and would cause leakage — they were used only to construct the churn label itself.

4. CHURN PREDICTION: APPROACH AND RESULTS

We trained two models on the 13,727 existing members (excluding the 2018 new-enrollee group described above): a Logistic Regression baseline and a Random Forest. Both were trained with class balancing, since churned members (16.3%) are a minority class, and evaluated on a held-out 20% test set.

Model	ROC-AUC	Churned: Precision	Churned: Recall	Overall Accuracy
Logistic Regression	0.861	0.68	0.7	0.9
Random Forest (selected)	0.889	0.67	0.7	0.9

We selected the Random Forest as the production model. A ROC-AUC of 0.89 is a strong result for behavioral churn prediction on real-world data. In practical terms: the model correctly identifies 70% of members who will actually disengage, and when it flags someone as at-risk, it is correct roughly two-thirds of the time. This balance is appropriate for a retention use case — missing too many at-risk members is costly, but so is sending win-back offers to members who were never going to leave.

Rank	Feature	Importance	What it means
1	Months since last flight (2017)	22.50%	By far the strongest signal — recency dominates.
2	Total flights in Fall 2017	12.70%	Recent seasonal activity matters.
3	Active months in 2017	8.60%	Consistency of engagement, not just total volume.
4	Average monthly flights (2017)	8.20%	Overall flying frequency.
5	Total distance flown (2017)	6.80%	Volume of travel, correlated with frequency.

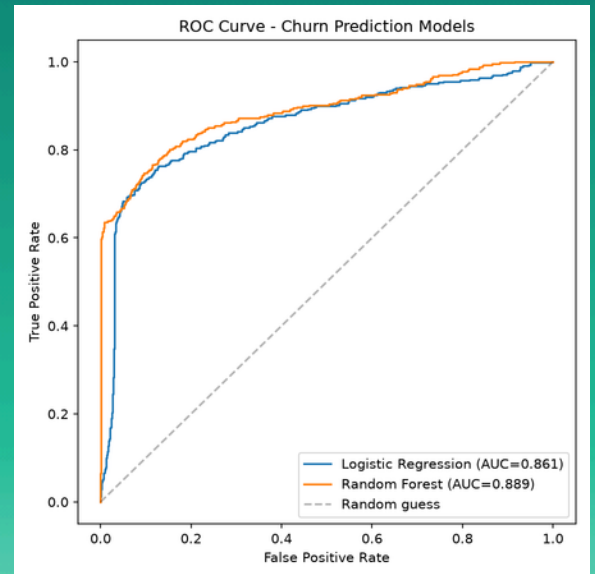


Figure: ROC Curve — Logistic Regression vs. Random Forest

4.1 What Actually Predicts Churn?

The single most important finding from the model is that behavior overwhelmingly outweighs demographics. Education, marital status, province, and loyalty card tier do not even appear in the top 15 most important features. Instead, the model is driven almost entirely by how recently and how often a member has been flying.

The practical implication is significant: an airline does not need a complex demographic profile to spot a member who is drifting away. A simple recency-based alert — “this member hasn't flown in N months” — already captures most of the signal, and is something the marketing team can act on immediately, even before a sophisticated model is deployed.

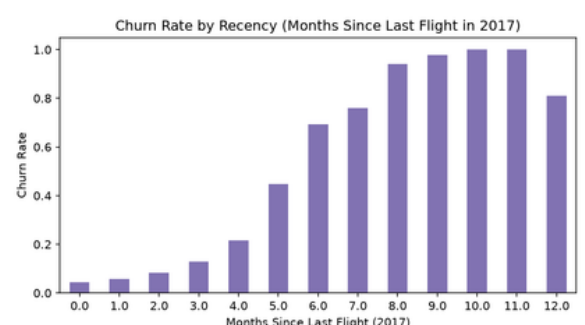


Figure: Churn Rate by Recency (Months Since Last Flight, 2017)

5. CUSTOMER SEGMENTATION AND VALUE

The brief asked us to interrogate whether CLV tells the full story about customer value. To answer this, we clustered all 16,737 members using K-Means on behavioral and value features — flight frequency, distance, points activity, recency, tenure, and CLV — deliberately excluding demographics from the clustering itself. We then profiled the resulting groups using demographics and churn risk, to see how these signals interact.

We tested cluster counts from 2 to 8 using silhouette scores. While k=2 scored highest, it only separated “active” from “inactive” members — too coarse to be useful for marketing. We selected k=4 as the best balance of statistical validity and business interpretability, then split one resulting cluster further (see below), arriving at five final segments.

Segment	Size	Churn Rate	Avg. CLV	Avg. Flights/Year	Persona
Loyal Frequent Flyers	4,556	4.00%	\$8,075	24.8	Highest activity, flew very recently, longest
New Members (2018)	2,767	n/a	\$8,112	n/a	Just joined — no behavioral history yet
Lapsed / Critical Risk	2,314	64.30%	\$8,112	0.6	Existing members gone quiet — 8+ months since last flight
Engaged Redeemers	2,130	7.40%	\$7,815	15.5	Moderate flyers who actively redeem points
Point Hoarders	4,970	8.60%	\$7,859	14.9	Moderate flyers, low churn, but barely redeem ~22,000

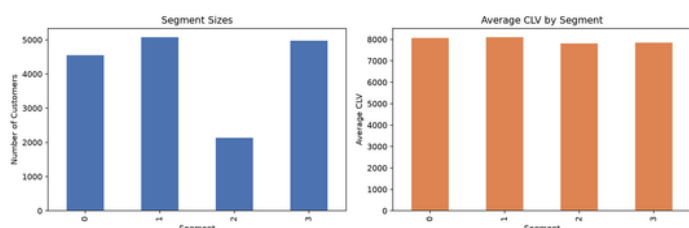


Figure: Segment Sizes and Average CLV by Segment

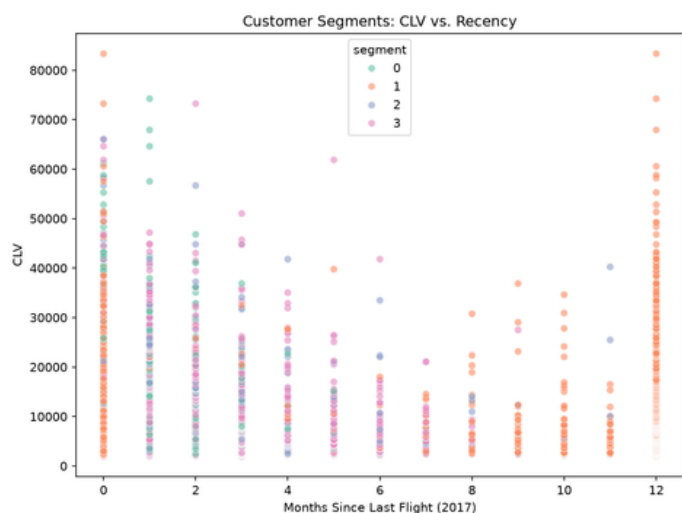


Figure: Customer Segments: CLV vs. Recency (scatter plot)

5.1 The Headline Finding: CLV Does Not Reflect Engagement or Risk

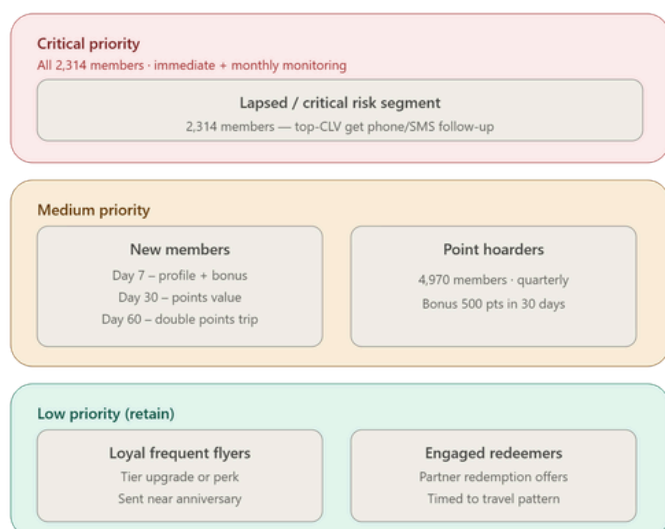
Every segment has an almost identical average CLV — between \$7,815 and \$8,112 — despite representing dramatically different customers. A “Loyal Frequent Flyer” who flies nearly 25 times a year and a “Lapsed/Critical” member who has flown almost nothing for over eight months have essentially the same CLV, yet one has a 4% chance of leaving and the other a 64% chance. This means that today, if the marketing team sorted their customer list by CLV alone, they would have no way of distinguishing a thriving relationship from one that is about to end. CLV, as currently calculated, answers “how valuable has this customer historically been,” but not “how likely are they to remain valuable.” The segmentation and churn model together close this gap.

5.2 Why “New Members” and “Lapsed/Critical” Needed to Be Separated

Initial clustering grouped 2018 new enrollees together with genuinely lapsed existing members, because both groups show near-zero 2017 flight activity. However, these are fundamentally different populations requiring opposite messaging: a brand-new member needs onboarding and encouragement, while a long-standing member who has gone quiet needs a win-back offer that acknowledges their history. We split this group using enrollment year, producing the five segments shown above. This is a good example of why segmentation needs to be checked against business logic, not just statistical separation.

6. SMART RETENTION:

Each segment below is mapped to a specific, executable action — designed so an operations team could pick this table up tomorrow without further interpretation. Priority levels (Critical, Medium, Low) are encoded directly into the dashboard, alongside each member's individual model-predicted churn risk score, so the highest-value interventions are surfaced first.



This logic is implemented directly in the working prototype (Section 9):

every member record carries a segment, a priority level, a model-predicted churn risk percentage, and the specific recommended action — so the marketing team's workflow becomes “filter by priority, work down the list.”

7. THE MOST INTERESTING THINGS WE FOUND

1. CLV is disconnected from engagement and risk. All five segments have nearly identical average CLV (within a \$300 band), despite churn rates ranging from 4% to 64%. CLV alone cannot be used to prioritize retention spend.

2. Recency beats everything — including demographics. “Months since last flight” alone accounts for over a fifth of the model's predictive power, more than all demographic variables combined. A simple recency alert is a powerful, low-tech early warning system.

3. The largest segment (Point Hoarders, 4,970 members — nearly 30% of the base) is quietly loyal but disengaged from the rewards mechanism itself — they fly regularly but barely redeem the ~22,000 points/year they accumulate. This is a latent liability and a missed engagement opportunity.

4. “Inactive-looking” members are not one group. Two populations — brand-new 2018 enrollees and long-standing members who went quiet — look identical on raw 2017 activity data (both near zero), but require opposite retention messages. Segmenting by enrollment history was essential to avoid a one-size-fits-none campaign

5. The Lapsed/Critical segment (2,314 members, 64.3% churn) represents a concentrated, addressable pocket of risk — roughly \$18.5 million in CLV is associated with this group alone ($2,314 \times \sim \$8,000$ average CLV), making it a natural first target for a focused win-back budget.

8. THREE BUSINESS RECOMMENDATIONS

The following recommendations are designed to be relevant to both a CFO (cost, risk, and revenue protection) and a CMO (customer experience and campaign design).

8.1 Recommendation 1: Replace “CLV-only” Targeting with a Combined Value + Engagement Score

Because CLV does not currently distinguish a thriving customer from one about to leave, we recommend introducing a simple second metric — an engagement or recency score — alongside CLV in all marketing and finance reporting. Pairing the two lets the business identify its true priority group: high CLV combined with declining engagement. For the CFO, this protects revenue that is currently invisible to risk monitoring. For the CMO, it turns a single “value” number into a two-dimensional map of who to nurture, who to reward, and who needs urgent attention.

8.2 Recommendation 2: Launch a Time-Boxed Win-Back Campaign for the Lapsed/Critical Segment

This segment of 2,314 existing members has a 64.3% churn rate and collectively represents an estimated \$18–19 million in CLV. We recommend a 60-day, budgeted win-back campaign: an automated bonus-points offer for the full segment, with personal outreach (phone or SMS) reserved for the highest-CLV individuals within it — a list the dashboard already produces. For the CFO, this is a bounded, measurable investment against a quantified at-risk revenue pool. For the CMO, it is a ready-made, segment-specific campaign brief with no further targeting work required.

8.3 Recommendation 3: Activate the “Point Hoarder” Segment Before Unused Points Become a Liability

Nearly 30% of the customer base — 4,970 members — accumulates points at a healthy rate but redeems almost none of them. Unredeemed points sit on the balance sheet as a future liability and do not reinforce the emotional connection that drives loyalty. We recommend a recurring “use your points” campaign for this segment, paired with a review of points-expiry policy. For the CFO, this manages a growing liability proactively rather than reactively. For the CMO, redemption activity is one of the strongest drivers of continued engagement — converting hoarders into redeemers is likely to reduce their (currently low, but not zero) churn risk further.

9. WORKING PROTOTYPE: THE RETENTION DASHBOARD

All of the analysis above is operationalized in a Streamlit dashboard designed so that a first-time, non-technical user can identify who needs attention and what to do about it without reading a manual. The dashboard loads the full scored and segmented customer base and provides:

- Top-line metrics: total members, count of Critical-priority members, average CLV, and count of high churn-risk members (model score $\geq 50\%$).
- Filterable views by priority level, segment, and model-predicted risk level.
- A “Recommended Actions by Segment” table — the retention matrix from Section 6, with live customer counts and average churn risk per segment.
- A “Top 20 Highest Churn Risk Customers” table — a ready-made daily/weekly call list for the retention team.
- A customer lookup tool: enter a Loyalty Number to see that member's segment, CLV, priority, churn risk, and recommended action in plain language.

PROTOTYPE LINK :



<https://airline-loyalty-project-snvj4uwmy3ir8pevzy5tyw.streamlit.app>

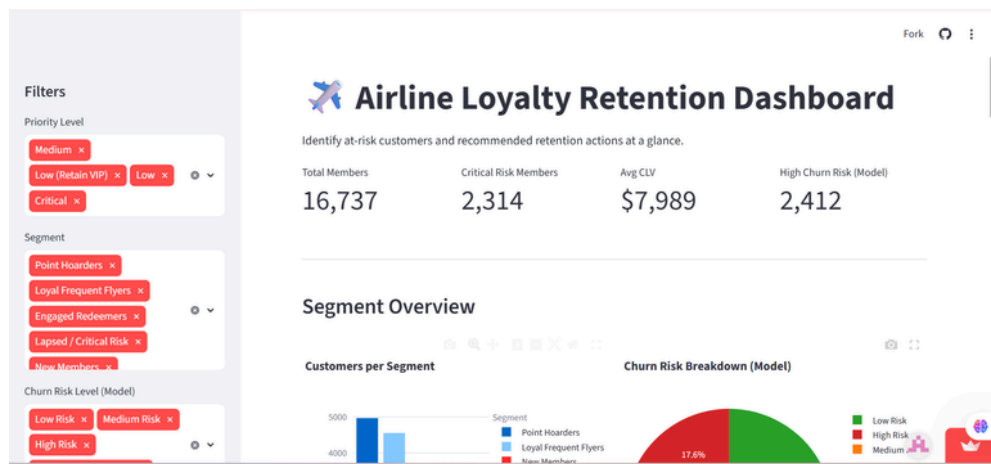


Figure: Dashboard — Top-line Metrics and Segment Overview

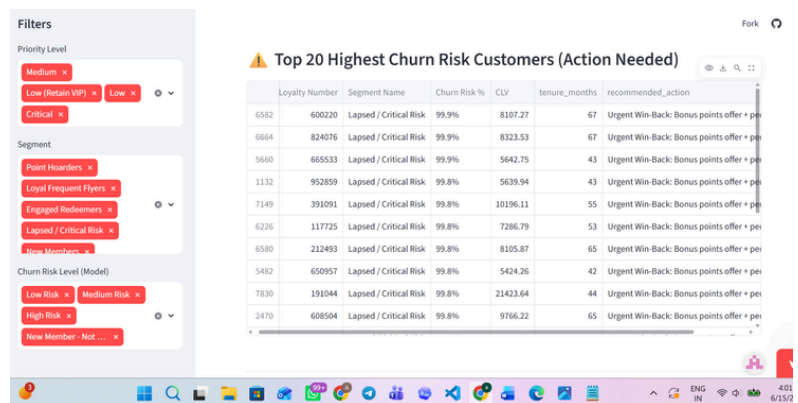


Figure: Dashboard — Top 20 Highest Churn Risk Customers

Recommended Actions by Segment

	Segment Name	priority	recommended_action	Customer_Count
0	Engaged Redeemers	Low	Cross-Sell: Partner redemption offers (quarterly)	21
1	Lapsed / Critical Risk	Critical	Urgent Win-Back: Bonus points offer + personal outreach	23
2	Loyal Frequent Flyers	Low (Retain VIP)	Loyalty Reward: Tier upgrade / exclusive perk (protect top customer)	45
3	New Members	Medium	Onboarding Series: Day 7/30/60 engagement nudges	27
4	Point Hoarders	Medium	Points Activation: "Use it or lose it" redemption nudge	45

Figure: Dashboard — Recommended Actions by Segment

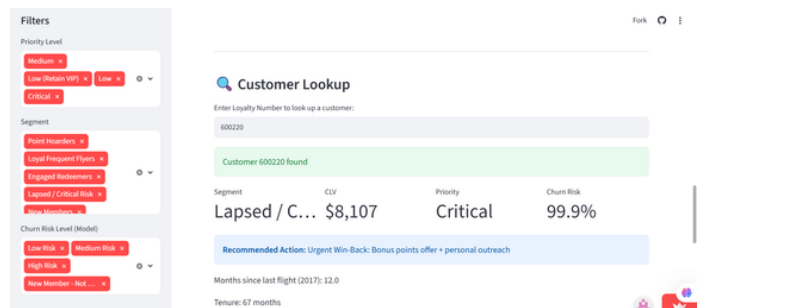


Figure: Dashboard — Customer Lookup and Full Filtered List

10. ASSUMPTIONS, LIMITATIONS, AND FUTURE WORK

- The behavioral dataset only covers 2017–2018 (24 months), which constrained the prediction framework to a single feature/outcome split. With a longer history, the model could be validated across multiple yearly cycles rather than one.
- The relationship between recorded “cancellation” dates and continued post-cancellation flight activity (Section 3.5) is not fully explained by the data alone and should be clarified with the data owners before this churn definition is used in production.
- CLV itself was treated as given; we did not re-derive it. Our finding is that, as currently defined, it does not vary meaningfully with engagement — we recommend the finance team review its underlying formula as a follow-up.
- As more data accumulates, the model should be retrained periodically (e.g., quarterly) and its precision/recall trade-off revisited as the business's tolerance for false positives (unnecessary retention offers) versus false negatives (missed at-risk members) becomes clearer.